CPSC 8440: Generative AI: Technical and Human Perspectives

Module 1.1: Introduction to this Course

Why Generative AI?

- Generative AI is transforming content creation and real-world decision-making
- Generative AI capabilities emerge from model architectures and data
- Its real-world impact depends on trust, fairness, safety, and human use
- This course bridges **technical foundations** with **human-centered design and ethics**

Course Content Structure

- **Module 1-3: Large Language Models (LLMs)**
  - The architecture, training, and evaluation of LLMs
- **Module 4: Image Generation**
  - Various cutting-edge image generation models, from GANs to diffusion models
- **Module 5-8: Human-Centered Generative AI**
  - Human-centered approaches to generative AI, enhancing human well-being, accountability, and alignment with societal values

Course Learning Objectives

- Explain the core principles and architectures of generative AI
- Design and implement generative AI solutions for text, image, and multimodal applications
- Analyze and evaluate generative AI systems using quantitative and qualitative criteria
- Apply human-centered design to improve usability, transparency, and user trust in generative AI systems
- Critically assess the societal and cultural impacts of generative AI on creativity, communication, and policy

Instructors

- Dr. Siyu Huang
- Assistant Prof., School of Computing, Visual Computing Division
- Research Interests: Generative AI, Image Synthesis, Deep Learning

- Dr. Katie Park
- Assistant Prof., School of Computing, Human-Centered Computing Division
- Research Interests: Human-Computer Interaction, Human-Centered AI, Online Safety

Teaching Assistant

- Huaye Li
- Ph.D. student (Computer Science)
- Research Interests: Human-Centered AI, Cybersecurity

Office Hours

- Instructors: Tuesdays 2-3 PM on Zoom
- TA: Fridays 1-2 PM on Zoom

Grading

- **Assignments** (Coding, Reading/Writing): 40%
- **Quizzes**: 10%
- **Mid-term** (Module 1-3): 25%
- **Final exam** (Module 1-8): 25%

- Late work will be accepted for 5 days, with 20% being taken off the score for each day late

AI Policy

- The use of artificial intelligence (AI) tools is permitted in certain circumstances in this course
- Students will be informed in writing as to when, where, and how these tools will be permitted
- Responsible use of AI tools must be documented and cited
- Outside of the permitted circumstances, AI tools should not be used for your work

What is next?

- Review Syllabus
- Start Module 1
- Begin readings and activities